CHAPTER 45
CAPITAL INVESTMENT DECISION ANALYSIS

Introduction

Investment leads to the increase in the stock of capital. The capital is an important factor of production in the modern economy. But capital has special features which distinguish it from other factors of production. The demand for other factors such as labour depends on their marginal revenue product (*MRP*) in a given period and the price that has to be paid for it. However, capital is a durable factor which can last and go on contributing to the production of goods for several years in future.

Firms sometimes take capital on rent per period for its use just as they hire workers for a monthly wage. But quite frequently the firms directly purchase the capital assets such as plant, factory buildings, machines and incur a lot of investment expenditure. When a firm undertakes investment expenditure for buying a machine or building a factory it has to compare the cost it has to incur *now* (*i.e.* in the present) with the additional profits or revenue it will generate for a number of years in the future. Therefore, while purchasing a capital asset a firm has to estimate how the stream of future profits generated by it is worth today. This problem is not faced while deciding about hiring of labour or purchasing of raw materials. This is because in deciding about hiring labour or purchasing raw materials the firm merely compares their current marginal revenue product (*MRP*) with the wage of workers or the price of raw material, it has to pay. In other words, the element of time does not figure in the calculations by a firm when it hires or purchases labour or raw materials.

Since a capital asset goes on yielding profits or revenue for a firm for several years from now, the firm has to find the present value of the stream of cash flows in the form of profits or revenue and compare it with the cost it has to incur on purchasing or building it. Since in the calculation of the net present value of capital asset, rate of interest plays a crucial role, we will also be discussing how rate of interest is determined. Besides, since future flows of profits may be higher or lower than expected, the firms have to take into account the risk and uncertainty involved while making investment in a particular capital asset.

Stocks Versus Flows

Investment, as mentioned above, is addition to the stock of capital. Measurement of capital as a factor of production is different from that of other factors of production. *Capital is measured as a stock* such as the amount of machines, equipment etc. a firm possesses for production, while other factors of production such as labour and raw materials are measured as *flows* such as the number of labour hours used for production per week, per month etc. It may also be noted that *output of the firm is also measured in terms of flows*, that is, output per week, per month or per year.

Consider the case of a producer who manufactures ceiling fans and has set up a factory equipped with required machines and other capital goods worth Rs 5 crores. He uses labour, copper and steel which are the variable factors to use them along with the capital stock to manufacture ceiling fans. The output of ceiling fans per month is 1000 ceiling fans which are sold at Rs 800 per fan with total sales proceeds per month equal to Rs 800,000. Suppose Rs 250 is the variable cost per fan and therefore total variable cost of 1000 fans per month amounts to Rs. 2,50,000. Thus Rs. 5,50,000 are the

profits made by the producer per month for several months in future. Thus, profits of Rs. 5,50,000 (*i.e.* 8,00,000 – 2,50,000) are also flows which will accrue to the producer per month in future. Thus, capital stock of Rs 5 crores yields to the firm the flow of profit of Rs 5,50,000 per month.

Now an important question is whether investment made to build the capital stock worth Rs 5 crores to yield monthly profits of Rs. 5,50,000 in future is economically sound decision. To examine this we have to convert the future cash flows of profits into their present value by using market rate of interest and then compare it with the cost of capital built up. If the present value of the flows of money profits for several months in future exceeds the cost of capital stock built up, the investment made will be profitable and would therefore increase the wealth of the firm.

On the other hand, if the present value of future cash flows of profits calculated by discounting at the market rate of interest falls short of the cost of capital investment, it will not pay the firm to undertake to invest in the capital stock. This is because the same monetary funds can be invested in bank's fixed deposits or in purchasing bonds or other interest-bearing financial assets to earn market rate of interest.

It follows from above that evaluating profitability of capital investment in projects we employ the concept of discounted value of the stream of cash flows of profits generated by that capital investment. In what follows we first explain the way in which present discounted value of future cash flows is obtained and then discuss the present value criterion of evaluating a capital investment. We will then compare this criterion with the alternative investment criterion of *Internal Rate of Return* (IRR).

Discounting and Present Values

People prefer present consumption to future consumption. Therefore, Rs. 100 at hand today is not worth the same as Rs. 100 to be received a year after. In other words, people prefer Rs 100 today to Rs. 100 to be received after a year. In evaluating investment projects which yield returns or cash flows over a number of years in future, it is imperative to know how much Rs. 100 paid in the future is worth today. In other words, what is the present value of Rs. 100 to be received after one year, after two years, after three years and so forth. How much cash flow of Rs. 100 to be received in a future year is worth today depends on the interest rate at which one can borrow or lend money. Therefore, a future cash flow is discounted at the interest rate to obtain its *present discounted value* (*PDV*) or what is simply called *present value* (*PV*). Suppose you have given a loan of Rs. 100 today to some body at 10 per cent rate of interest. Then after one year you have will receive Rs. 110, that is, $100+100\times\frac{10}{100}$ or $\left(100+100\times\frac{1}{10}\right) = 100\,(1+0.1) = 110$. This means Rs. 100 today is equivalent to Rs. 110 a year after, if rate of interest happens to be 10 per cent per annum. In other words, Rs. 110 after a year is worth Rs. 100 today, given 10 per cent rate of interest. Note that in obtaining present discounted values, *interest rate is expressed in proportionate decimal form*, that is, 10% = 0.10. The present value can be found as under

$$100\,(1+0.10)=110$$

$$\text{or} \qquad 100 = \frac{110}{1+0.10} \qquad \ldots (i)$$

The equation (*i*) shows if Rs. 110 to be paid after a year is discounted at 10 per cent interest rate, we obtain its present value of Rs. 100. To put in general terms, if r represents rate of interest, then present value (*PV*) of Rs. 110 to be paid after a year is

$$PV = \frac{110}{1+r}$$

Similarly, if Rs. 110 is to be paid after two years, its present value,

$$PV = \frac{110}{(1+r)^2}$$

Further, if Rs. 110 is to be paid after three years, its present discounted value,

$$PV = \frac{110}{(1+r)^3}$$

and so forth for further future years of payment.

The concept of present discounted value is extensively applied and used in financial matters and to facilitate accounting a Table has been prepared which provide present discounted value (*PDV*) of Re. 1 paid after 1 year, 2 years, 3 years, 5 years and so on and this table of present discounted value (*PDV*) is given in Log Tables. We reproduce it in Table 45.1 so that the students should learn to use them in actual exercises. Thus

Table 45.1. PDV of Re. 1 paid in the Future

Interest Rate	1 Year	2 Year	5 Year	10 Year	20 Year
0.01	$0.990	$0.980	$0.951	$0.905	$0.820
0.02	0.980	0.961	0.906	0.820	0.673
0.03	0.971	0.943	0.863	0.744	0.554
0.04	0.962	0.925	0.822	0.676	0.456
0.05	0.952	0.907	0.784	0.614	0.377
0.06	0.943	0.890	0.747	0.558	0.312
0.07	0.935	0.873	0.713	0.508	0.258
0.08	0.926	0.857	0.681	0.463	0.215
0.09	0.917	0.842	0.650	0.422	0.178
0.10	0.909	0.826	0.621	0.386	0.149
0.15	0.870	0.756	0.497	0.247	0.061
0.20	0.833	0.694	0.402	0.162	0.026

If rate of interest (r) *is 8 per cent per annum*, then

$$PDV \text{ of Re 1 paid after 1 years} = \frac{1}{1+r} = \frac{1}{1+0.8} = 0.926$$

$$PDV \text{ of Re 1 paid after 2 years} = \frac{1}{(1+r)^2} = \frac{1}{(1+0.8)^2} = 0.857$$

$$PDV \text{ of Re 1 paid after 5 years} = \frac{1}{(1+r)^5} = \frac{1}{(1+0.8)^5} = 0.681$$

If present discounted value of a large number of rupees is to be found we just multiply that number by the *PDV* of Re 1 paid after the given number of years. Thus, if we have to find the present discounted value of Rs. 5000 to be paid after two years with the given rate of interest equal to 8 per cent per annum we can obtain it as under

$$PDV \text{ or } PV \text{ of Rs 5000 to be paid after 2 years} = 5000 \times \frac{1}{(1+0.8)^2} = 5000 \times 0.857$$

$$= \text{Rs. } 4285$$

Present Value of a Stream of Payments

Quite often investment projects yield a stream of payments or cash flows today and future years. Present value (*PV*) in this case can be calculated by adding the present discounted values of

the various amounts paid in various years discounted by the given rate of interest. Two types of payment streams are given in Table 45.2 and we have to find out the present value of these streams of cash flows if rate of interest is 10 per cent.

Table 45.2. *Two Payment Streams*

	Today	1 year	2 years
Payment Stream A :	Rs 500	Rs 500	0
Payment Stream B :	Rs 50	500	500

We have first to find present discounted value (*PDV*) of Re 1 paid in the future by consulting Table 45.1. We present below the present discounted value of the above two payment streams

$$PDV \text{ of stream } A = 500 + \frac{500}{1+r} = 500 + \frac{500}{1+0.10}$$

$$= 500 + \frac{500}{1+0.10} = 500 + 500 \times 0.909$$

$$= 500 + 454.5 = \text{Rs } 954.5$$

$$PDV \text{ of stream } B = 50 + \frac{500}{1+0.10} + \frac{500}{(1+0.10)^2}$$

$$= 50 + (500 \times 0.909) + (500 \times 0.826)$$

$$= 50 + 454.5 + 413.0 = \text{Rs } 917.5$$

From above it follows that the first stream *A* is preferred to the second stream *B* at 10 per cent rate of interest because the present value of stream *A* is greater than that of stream *B*.

Now, if rate of interest is 5 percent per annum, then

$$PDV \text{ of stream } A \text{ at 5 per cent interest rate} = 500 + \frac{500}{1+0.05}$$

$$= 500 + 476 = \text{Rs. } 976$$

$$PDV \text{ of stream } B \text{ at 5 per cent rate of interest} = 50 + \frac{500}{1+0.05} + \frac{500}{1+(0.05)^2}$$

$$= 50 + 476 + 453.5 = \text{Rs. } 979.5$$

Thus, at 5 per cent rate of interest, present discounted value (*PDV*) of stream *B* exceeds that of stream *A*. Thus, at 5 percent interest rate, stream *B* is a preferred payment stream. It therefore follows which of the payment stream would a businessman would prefer to receive depend on rate of interest. Besides, it follows from above as *interest rate rises, the present value of a given income stream falls*.

Value of a Bond

A bond is a financial instrument through which corporate firms and government borrow money from the public. The borrower, corporate firm or government, which issues bond agrees to pay the bond holder, that is, who purchases the bond, a stream of money (*i.e.* cash flows). For example, the Indian government's 10 years bond of Rs. 1000 might make a coupon payments of Rs. 100 per year for the next 10 years and then after the expiry of 10 years, the principal amount is paid back. Now, a significant question is what is the value of a bond to you? or ,in other words, how much bond is worth to you today? To know how much bond is worth today which determines how much an

individual would pay for the bond, we have to calculate the present value of its payment stream. Thus

$$PDV = \frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \cdots + \frac{100}{(1+r)^{10}} + \frac{1000}{(1+r)^{10}}$$

where r is the rate of interest.

As shown above, the present value of a payment stream depends on rate of interest. *As the rate of interest falls, present value (PDV) of a bond with a given payments stream increases. On the other hand, present value of a bond decreases as the interest rate increases*. In Figure 45.1 we have shown a curve of *PDV* of cash flows (in thousand) at different rates of interest. For example, by applying the above formula we calculate the Present Discounted Value (*PDV*) of cash flows of a 10 year bond paying Rs 100 annually on a principal of Rs. 1000 to be equal to Rs. 1386 when the interest rate is 5 per cent.

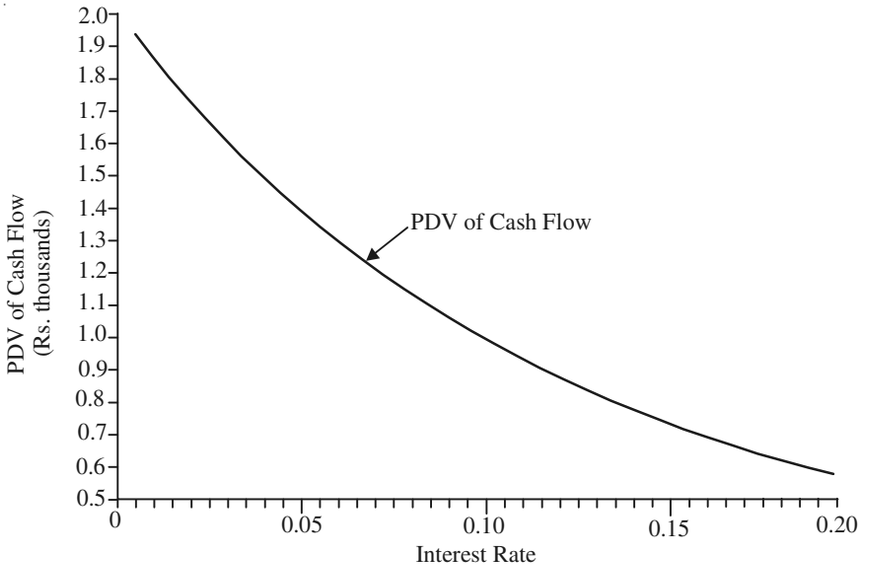


Fig. 45.1. *Present Value of the Cash Flow from a Bond*

Thus when the interest rate is 15 per cent per annum, on calculation we find that the same bond is worth (*i.e.* present value) only Rs. 749.

Perpetuity

A perpetuity or a perpetual bond is a special type of a bond that yields a fixed amount of cash flow annually for ever in future. The present value of perpetual cash flow (*i.e.* payment stream) of Rs. 100 per annum is given by the following infinite summation :

$$PV = \frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \frac{100}{(1+r)^3} + \frac{100}{(1+r)^4} + \cdots$$

$$PV = 100\left(\frac{1}{(1+r)} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \cdots\right) \quad \ldots (ii)$$

The above present value of the stream of cash flows paid annually can be easily obtained by using the following formula which gives the present discounted value of cash flow of Re 1 per annum for ever.*

$$PV \text{ of Re. 1 payable per annum for ever} = \frac{1}{r}$$

where r is rate of interest.

Thus, Present Value (PV) of the perpetuity of Rs 100 = $\frac{100}{r}$

If rate of interest is 5 per cent, then PV of a perpetual bond with Rs 100 as annual cash flow

$$= \frac{100}{0.5} = \text{Rs } 2000$$

If $r = 10$ per cent, then

PV of the above bond $= \frac{100}{0.10} = \text{Rs } 1000$

Thus the *lower the rate of interest, the higher the present value of a perpetual bond.*

The Effective Yield (i.e .Rate of Return) on a Bond

Bonds of many corporate firms and government are often traded in the bond market and reported by the media. Therefore, what these bonds are worth (that is, their present value) can be known directly by taking a glance at the prices of bonds traded in the market. Though bond prices can be known easily but to determine the effective yield or rate of return on them is bit difficult. *The rate of interest which equates the present discounted value of the bond's payment stream of cash flows with the bond's market price is the effective yield or rate of return on the bond.* In this way the effective yields or rates of return on various bonds can be calculated. We can compare them with the rates of return on other available investment opportunities for the investors.

In the previous section we have explained the equations that show how the present value of bonds depend on the interest rate used to discount the bond's payment stream. These equations can be rewritten so as to determine effective yield from the bond's value. It is easy to determine interest rate (*i.e.* effective yield) in case of perpetuity. Thus

PV of a perpetuity with Rs. 100 as the annual money payment = $\frac{100}{r}$

If market price of a bond (*i.e. PV* of bond) is Rs 1000, then rate of interest r or effective yield can

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*This can be proved as follows

According to a well-known mathematical theorem, the sum of an infinite series such as $x + x^2 + x^3 + x^4 + \cdots$ is equal to $\frac{x}{1-x}$ provided the absolute value of x is less than one. As $\frac{1}{1+r}$ is less than one, $\frac{1}{1+r}$ in equation (*i*) can be considered as x, then the sum of series in equation (*ii*) above can be written as

$$\frac{1}{1+r} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \cdots = \frac{\frac{1}{1+r}}{1-\frac{1}{1+r}}$$

$$= \frac{\frac{1}{1+r}}{\frac{r}{1+r}} = \frac{1}{1+r} \times \frac{1+r}{r} = \frac{1}{r}$$

With this the equation (*ii*) can be written as

$$PV = 100\left(\frac{1}{r}\right) = \frac{100}{r}$$

be obtained as under :

$$\text{Rs. } 1000 = \frac{100}{r}$$

$$r = \frac{100}{1000} = 0.10 \text{ or } 10 \text{ per cent}$$

Thus 10 per cent is the effective yield of the perpetuity in question.

To find out effective yield of 10 year bond with coupon payment of Rs 100 per year as given in the previous section is a difficult task as it involves a good deal of calculations. Rewriting the present value equation of 10 year bond with a terminal payment we have

$$PV = \frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \frac{100}{(1+r)^3} + \cdots + \frac{100}{(1+r)^{10}} + \frac{1000}{(1+r)^{10}}$$

Given the price of the 10 year bond (*i.e.* its present value *PV*), we can calculate the value of *r* consistent with bond's price. Some computer programs (even programs on scientific calculators) are available which can greatly help in calculating the rate of interest *r* consistent with the given bond price. The calculation can also be done by using the table of "*PDV* of Re 1 paid in the future" that is, the value of $\frac{1}{1+r}$, $\frac{1}{(1+r)^2}$, $\frac{1}{(1+r)^3}$ etc. as explained above. In Figure 45.2 below we have drawn a curve showing interest rate (*i.e.* effective yield) consistent with present value (or market price) of a bond. It will be seen from Fig.45.2 that if the price of a bond (*i.e. PV*) is Rs 1300, the effective

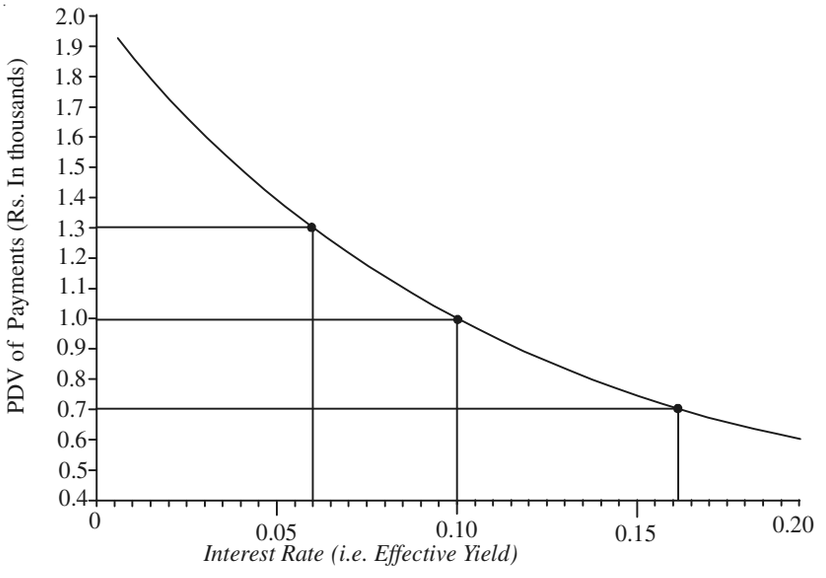


Fig. 45.2. *Effective Yield on a Bond*

yield or rate of return from it is around 6 per cent per annum. If that bond sells at Rs 700, the effective yield from it will be about Rs 16 per cent per annum.

Effective Yield and Risk

It is a common knowledge that yields from different bonds differ considerably. What are the reasons for the differences in yields (*i.e.* rate of return) from different bonds. Bonds of corporate firms yield more than government bonds. And corporate bonds of some eminent companies such as

Reliance, Tata *yield less* than those of financially weak corporate firms. The most important reason for the differences in yields of bonds is that different bonds have different degree of risk. The Indian government is unlikely to default, that is, fails to make payments of interest on its bonds and the principal sum borrowed. Therefore, the bonds of Government of India carry less risk and therefore yields on them are relatiely less as compared to the bonds of private corporate firms which can default on interest or principal payments on their bonds. And the bonds issued by the financially strong and reputed private corporate companies such as Reliance and Tata are less likely to default and the yields on their bonds are less as compared to those of the financially weak companies. Thus more risky the investment bond is, the higher must be yield on it. As a matter of fact, the higher yield on a more risky investment is a form of risk premium carried by it. The investors would like to invest in more risky bonds only if they are compensated by higher return or yield. Thus riskier bonds have higher yields.

Bond Prices and Rate of Interest : Why does yield or return on bonds vary ?

Prices of both the corporate bonds and Government securities with a certain fixed coupon interest rate (that is, with a fixed annual coupon payments for a certain maturity period) charges in the bond market. The changes in bond prices affect the effective yield or rate return on them. For example, if the price of a bond of ₹ 1000 carrying 100 annual coupon cash payments, falls, say to ₹ 950, it implies that now purchase of the bond at price ₹ 950 will yield ₹ 100 per annum, that is, yield on that bond will now be more than 10 per cent. On the other hand, rise in bond price will cause yield on previously issued bond to fall.

Now the question is why bond prices change even though carrying a fixed coupon interest rate (i.e., fixed annual cash payment). Prices of previously issued bonds change due to changes in market rates of interest. There is inverse relation between bond prices and market rate of interest. When market rate of interest goes up, prices of previously issued bonds fall, and when market rate of interest falls, prices of old bonds must rise. An example will clarify this inverse relation between bond prices and changes in rate of interest. Suppose Reliance Industries issued a 10 year bond of ₹ 1000 in the year 2009 when rates of interest were low at 8% and thus promised to pay ₹ 80 per year on a bond of ₹ 1000 till the end of the 10 year maturity period (Besides, it will repay the original sum of ₹ 1000 after the expiry of maturity period). Now, suppose that in March 2011, market rate of interest rises to 9 per cent, then the newly issued 10 year bond will now be issued with 9 percent rate of interest (that is, yielding ₹ 90 annually on a 10 year bond of ₹ 1000). Therefore, in March 2011, no body will buy the previously issued 10 year bond of ₹ 1000 yielding ₹ 80 per annum as similar companies will now issue 10 year bond at 9 per cent coupon interest rate. Therefore, the old bonds must now yield at least 9 per cent if they are to be purchased in the secondary market. However, to yield 9 per cent return the price of previously issued 10-year bond must fall. If its price falls to ₹ 850 then the old 10-years bond of ₹ 1000 at 8% with coupon annual payment will yield a return of $\frac{80}{850} \times 100 = 9.4\%$ and therefore it will now be worth while to purchase it. Thus the price of old bonds of the same maturity period with 8% interest rate must be sold below its original price if it is to yield higher interest rate of at least 9 per cent.

Inverse Relationship between Bond Prices and Interest Rate

It follows from above that there is inverse relationship between changes in interest rate and bond prices. When the market interest rates in the economy rise, the prices of previously issued bonds at lower coupon interest rate must fall. Similarly, when market interest rates fall, the prices of previously issued similar bonds must rise. It is worth noting that interest rates in the economy often change because of changes in monetary policy of the Central Bank of a country. Depending on the economic situation, the Central Bank of the country sometimes raises interest rate and sometimes lowers interest rates. In addition to changes in interest rate, bond prices change due to changes in demand for investment in these bonds and other securities. Note that unlike equity (i.e, corporate

shares) bonds or securities are debt instruments. The investment in these bonds is made by wealthy individuals who do not want to take risk of investment in shares whose value can fall drastically resulting in negative return. Investment in these bonds are generally made by banks, insurance companies, mutual funds, pension funds and more recently FII (foreign institutional investors). The demand for bonds (debt instruments) changes which causes changes in demand prices. They compare risk adjusted rate of return in shares (equity capital) with yield on bonds and decide to allocate their investment resources in equity capital and debt capital to ensure a certain return on their investment. Similarly, business men, banks, mutual funds, FII want to keep a certain balance in investment between shares and bonds so as to ensure a certain rate of return on their capital investment. Besides, the changes in demand for investment in bonds depends on the changes in economic environment such as changes in monetary policy by RBI, rate inflation in the economy, capital inflows by FII, Government borrowing programme etc. All these affect demand for bonds and their prices. The changes in bond prices, as explained above, cause changes in their effective yields.

It will be interesting to note that in the year 2010 in the US yield or interest rate on bonds and securities has been estimated to be near 2 per cent while in India yield on them is presently around 8 per cent. That is why FIIs of the US and other developed economies are pumping more funds into India to purchase equity shares and bonds. This is affecting share prices and bond prices in India.

CAPITAL INVESTMENT DECISION : NET PRESENT VALUE CRITERION

Our intertemporal analysis of investment made in the previous chapter and the concept of present values have an important practical application for decision making by a business firm. Managers of private business firms and public enterprises have often to decide which investment projects are to be undertaken. For example, business manager has often to decide whether he should build a new plant, purchase a new high-tech computer, should he scrap the old machine, how much to invest in research and development (R & D) or advertising and so forth. Crores of rupees may be invested in a building a factory and instaling machines and equipment in it. And this factory and equipment will last for several years to come and affect the profits of the firm. Besides, the future cash flows that a capital investment generates are quite uncertain. And once the project has been constructed, the firm cannot usually dismantle it and resell it to recover the investment cost incurred. Thus, a *good deal of investment cost incurred represents a sunk cost*. This requires a careful assessment of investment costs incurred and revenues expected before a decision is made whether to undertake a particular investment project. As investment projects yield return over a number of years, they involve a stream of cash flows (*i.e.* a sequence of incomes and costs). Our intertemporal analysis of lending and borrowing, saving and investment made in the previous chapter is very useful for examining the problem of these investment decisions. Whether or not to make investment in a project is based on some *basic decision rules* or what are called *investment criteria*.

Net Present Value Criteria or Rule for Capital Investment Decision : Choice of a Project

An investment project yields a return in the form of a stream of a cash flows and it also involves cost to be incurred in the present. For instance, for a project such as investment for purchasing a machine, building a factory, acquiring an educational or technical skill, an individual has to incur costs in the current year. Return from an investment project are obtained as cash flows of income or output in future years. An important decision rule for investment is the net present value criterion. The basis of this criterion or rule is that a unit of a commodity or a rupee in a future year is worth less than a unit of the commodity or rupee this year. Therefore, cash or output flows from a project accruing at different times cannot be simply added to determine whether or not to make investment in the project. The reason for this is that returns in future years have to be discounted at a rate of interest.

Let us first explain choice among investment projects when two periods are involved, that is, the present year and the next year. In an investment project, since in the present year cost is incurred,

cash flow is negative. The returns start from the next year in the form of increase in income or output, the cash flows are positive in the future years. In the choice of project a firm is guided by the *principle of maximisation of wealth.* Investment in a project which increases wealth is desirable. Further, as mentioned earlier, while making investment decision, the firm will calculate the net present value of the future cash flows that it expects to obtain from the investment and compare it with the cost of the investment project. The *net present value of a project is the value today of the net cash flows in the future years obtained by discounting them at the market rate of interest minus the cost incurred on the project in the current year*. Note that costs incurred on maintenance of a machine or any other type of project have to be substracted from receipts or income flows to obtain the net cash flows for the future years. A project which will yield income only from a year now and has no scrap value, net present value is given by

$$NPV = -C + \frac{R_1}{1+i}$$

where *NPV* denotes the net present value of a project, C_0 is the cost of the investment project in the current year and will therefore be negative, R_1 is the net cash flow in the next year and i is the market rate of interest.

If the net present value of an investment project is positive, this means it increases wealth of the business firm. As the objective of the firm is to maximise its wealth, *it will be desirable to undertake investment in the project if its net present value is positive (i.e. if NPV > 0)*. This is a decision rule for deciding about investment in a project.

Net Present value Rule for Capital Investment Decisions : Multiperiod Case Many investment projects yields a stream of returns (*i.e.* cash flows) beyond two periods. The net present value rule applies in this case too. However, we have to extend the net present value rule to deal with several years. Return or net cash flow of Rs. 1000 occurring in the third year is worth less than Rs. 1000 occurring in the second year. Likewise, Rs. 1000 accruing in the fourth year is less than Rs. 1000 accruing in the third year. Therefore, in a multiperiod case the net present value of a project can be obtained as under :

$$NPV = -C + \frac{R_1}{1+i} + \frac{R_2}{(1+i)^2} + \frac{R_3}{(1+i)^3} + ... + \frac{R_n}{(1+i)^n}$$

where *NPV* is the present value of the project, C is the cash flow during the current year (It is negative in case of an investment project as cost has to be incurred on a project in the current year) $R_1, R_2, R_3 ... R_n$ are the positive net cash flows in the form of income after first, second, third year and so on.

Net Present Value Criterion : *Approve or undertake any investment project whose net present value is positive and reject any investment project whose net present value is zero or negative.*

Now an important question is at what rate cash flows in future be discounted to arrive at net present value. The market rate of interest is not always the proper discount rate. The discount rate which ought to be used is determined by the alternative uses of the funds available to the firm. Instead of making investment in a particular project, another investment project may be there that generates a different stream of cash flows or profits. Alternatively, a firm may invest in a bond that yields a different rate of return. In fact, for a firm there is the opportunity cost of capital. The opportunity cost is the rate of return the a firm can earn by investing the available funds in an alternative similar project instead of investment in a particular project. By similar investment project we mean the one which involves the *same risk*. As explained earlier, the making investment in projects with higher risks would require greater return or yield. Therefore, "the opportunity cost of investing in a project is the return that one could earn from another project or asset of similar

riskiness"[1].

In a later section we will explain how a firm makes adjustment for risks involved in various investment projects. In the present analysis we assume that the investment project does not involve any risk, that is, the firm is quite certain about the stream of future profits π_1, π_2, π_3 etc. generated by an investment project. Besides, the opportunity cost used to discount the future stream of profits is the *risk-free return*, such as the return that can be earned from investing in government bonds. For example, if the investment project is expected to last for ten years, then the interest rate on a 10 year government bonds can be used to obtained the net present value (*NPV*) of a project. If the *NPV* of the project is positive (*i.e.* if $NPV > 0$), the yield from the project will exceed the opportunity cost and therefore it is worthwhile to make investment in the project.

The net present value criterion for investment assumes that there is no uncertainty about costs to be incurred and the stream of profits (cash flows) in future. The alternative investment criterion such as the internal rate of return criterion and the profitability index criterion have been proposed. But, as we will discuss below, the *NPV* criterion is better than these other investment criteria.

NUMERICAL EXAMPLES

Example 1. *If an investment project costs Rs. 10,000, market rate of interest is 15 per cent per annum and it yields net return of Rs. 16, 100 in the next year. Should the investment in the project be made ?*

Sol. The net present value of the this project is given by

$$NPV = -C_0 + \frac{R_1}{1+i}$$

$$NPV = -10,000 + \frac{16,100}{1+0.15}$$

$$= -10,000 + \frac{16,100}{1.15}$$

$$= -10,000 + 16,100 \times \frac{100}{115}$$

$$= -10,000 + 14,000 = Rs.\ 4,000$$

Thus, undertaking of this project well increase wealth of the firm by Rs. 4000. Therefore, investment in this project should be made.

Take another example.

Example 2. *Suppose a firm is planning to purchase a machine which costs Rs. 20,000 and interest is 10 per cent per annum and anticipated yield from it next year is Rs. 22,000 and at the end of the next year the machine has no salvage or scrap value. Is it worthwhile to approve the project.*

Sol. Net present value of the machine

$$NPV = -C_0 + \frac{R_1}{1+i} = -20,000 + \frac{22,000}{1+0.1}$$

$$= -20,000 + \frac{22,000}{1.1}$$

1. Robert S. Pindyck and Daniel L Rubinfeld, *Microeconomics,* 3rd ed. Prentice Hall of India, New Delhi, 1995, p-533.

$$= -20,000 + 22000 \times \frac{10}{11} = 0$$

Thus the net present value of the machine is zero. It is not worthwhile to make investment in the machine. However, it may be noted that if the yield from the machine next year exceeds Rs. 22,000, its net present value will become positive and it will then be desirable to purchase it for the production of a commodity. Note that if the yield next year remains Rs. 22,000 but rate of interest falls below 10 per cent, the net present value will again become positive.

Example 3. *Suppose a firm is considering to invest in a project. Its cost this year is Rs. one lakh and its net yield in year one from now is Rs. 1.25 lakh and its scrap value is zero. What is the highest rate of interest at which the project should be undertaken*

Sol. To determine the required value of the rate of interest (i), we should make the net present value equal to zero. Thus

$$NPV = -C_0 + \frac{R}{1+i} = 0$$

$$-100,000 + \frac{125,000}{1+i} = 0$$

$$\frac{125,000}{1+i} = 100,000$$

or $$100,000(1+i) = 125,000$$

$$1+i = \frac{1,25,000}{100,000} = \frac{5}{4}$$

$$i = \frac{5}{4} - 1 = 0.25$$

This means if the rate of interest is 0.25, that is, 25 per cent, the net present value of project is zero. Thus, if the rate of interest falls below 25 per cent, the net present value will become positive. Thus the project should be undertaken for any value of the rate of interest below 25 per cent.

Example 4 : *Suppose a firm is considering to buy a machine today for Rs. 25,000. The use of this machine in the production process of a commodity causes firm's net income to rise by Rs. 15000 in each of the next two years. Assume the rate of interest is 10 per cent and machines has no scrap value. Is it desirable for the firm to purchase the machine* ?

Sol.
$$NPV = -C_0 + \frac{R_1}{1+i} + \frac{R_2}{(1+i)^2}$$

$$= -25000 + \frac{15000}{1+0.1} + \frac{1500}{(1+0.1)^2}$$

$$= -25000 + 15000 \times \frac{10}{11} + 15000 \times \frac{10}{11} \times \frac{10}{11}$$

$$= -25000 + 13636.4 + 12396.7 = 1033.1$$

Thus we find net present value of the machine is positive and therefore it will increase the wealth of the firm by Rs. 1033. It is therefore desirable to make investment in it.

If the machine creates net revenue of Rs. 15000 in one more year, the net present value will further increase, making the machine more wealth generating. In case a machine yields net return in the third year also we extend our equation to the third year as well and calculate the net present value. Thus

$$NPV = -C_0 + \frac{R_1}{1+0.1} + \frac{R_2}{(1+0.1)^2} + \frac{R_3}{(1+0.1)^3}$$

Further, if at the end of two years the machine has some scrap value, say equal to Z, then value Z will also the discounted at the given market rate of interest. Thus, in this case

$$NPV = -C_0 + \frac{R_1}{1+0.1} + \frac{R_2}{(1+0.1)^2} + \frac{Z}{(1+0.1)^2}$$

Net Present Value Rule 2 : Choice among Multiple Investment Projects

In our above analysis we have been concerned with explaining a rule regarding whether or not a project should be approved for capital investment. Now, we will discuss the choice by a firm when several projects are available for investment. In this case we need to compare the net present values of the various projects. The project which has the highest present value should be undertaken. The project with the highest net present value will make greatest contribution to the objective of maximising wealth by a firm. Thus, in case of choice among multiple projects we arrive at the following rule. "*If the two or more mutually exclusive projects are available for investment, undertake the project which has the highest net present value.*"

It may be pointed out that we face difficulties in the use of this rule when projects, instead of being mutually exclusive are *interdependent*, that is, undertaking of a project may change the benefits or returns from the other projects. For instance, a project of flood control may increase the benefits or revenue from the project of growing wheat on an agricultural farm. In case of these interdependent projects we have to modify the rule. In this case we should compare not the present values of individual projects but various 'sets' of interdependent projects. Thus, for the choice of a single project from various projects, they must be mutually exclusive, that is, independent of each other.

Let us illustrate this decision rule concerning mutually exclusive multiple projects by giving an example.

Problem 5.

Two projects A and B are available to a firm. Cost in the current year and net cash flow in the next year of each is given below. Which project a firm should choose if market rate of interest is 10 per cent per annum.

Project	*Cost this year in the next year*	*Net cash flow*
Project A	100 lakh	120 lakh
Project B	150 lakh	180 lakh

Net present (NPV) of Project $A = -C_0 + \frac{R_1}{1+i}$

$$= -100 + \frac{120}{1+0.10} = -100 + 120 \times \frac{10}{11}$$

$$= -100 + 109.09 = 9.09$$

Net present value (NPV) of Project $B = -150 + \frac{180}{1.1}$

$$= -150 + 180 \times \frac{10}{11} = -150 + 163.63 = 13.63$$

Thus, net present value of Project B is greater than the net present value of Project A. Therefore, the firm which aims at maximising wealth should undertake Project B.

Present Value of an Annuity with Perpetual Constant Annual Yield

An interesting case is of annuity with a constant stream of net cash flows for ever (*i.e.* in perpetuity), starting from a year now. As explained above, the present value of such an annuity is given by

$$PV = \frac{R}{i}$$

where R is stream of payments which an investor receives, starting from the next year, i is the rate of interest. This can be proved as follows:-

$$PV = \frac{R}{1+i} + \frac{R}{(1+i)^2} + \frac{R}{(1+i)^3} + ...\text{indefinitly}$$

In the above R can be factored out. Therefore,

$$PV = R\left[\frac{1}{1+i} + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + ...\infty\right] = \frac{R}{i}$$

Let us provide an example.

Suppose a bond yields a sum of Rs. 6,000 per annum indefinitely, starting from a year now. Assuming rate of interest is 12 per cent, its present value is given by

$$PV = \frac{R}{i} = \frac{6000}{0.12} = \frac{6000}{12/100} = 50,000$$

This is because if Rs. 50,000 are invested at 12 per cent rate of interest, it will yield Rs. 6,000 every year for ever $\left(50,000 \times \frac{12}{100} = 6000\right)$

Effect of Change in Rate of Interest on Net Present Value of an Investment Project

It is important to note that the rate of interest has an important bearing on the present value of an investment project. *The lower the rate of interest, the higher the present value of a project and vice versa.* Thus, in our above example of an annuity, with a constant annual yield of Rs. 6,000 for ever, if the rate of interest falls to 10 per cent, its present value will be

$$NPV = \frac{R}{i} = \frac{6000}{0.10} = 6000 \times \frac{100}{10} = 60,000$$

Thus, at a lower rate of interest the present value of the annuity has increased. This not only applies to the case of an annuity with a constant yield indefinitely but also to our earlier examples of present values of investment projects which yield cash flows for a limited number of years and also when annual yields are not constant.

It is also important to note here that cost of an investment project, (for example, purchasing a machine) may not be incurred entirely in the current period, some cost also occurs every year for its maintenance. Therefore, when finding out present value of a project, we need to consider *net cash flows* of the projects. In order to obtain these net cash flows from the extra revenue generated by a project, the cost incurred on it for its maintenance and operation has to be substracted from it. Further, as mentioned above, the projects have some scrap value, therefore when finding out present value of a project, the present value of its scrap value has also to be included in the stream of net cash flows of a machine or any other project.

INTERNAL RATE OF RETURN (IRR) METHOD

Another important method of evaluating investment project is internal rate of return method. This evaluation method is also based on present value of net cash flows generated by a project over its life period. *Internal Rate of Return (IRR) is the rate of discount that equates the present value of*

net cash flows equal to the initial investment cost of the project. If R_1, R_2, R_3...R_n represent the net cash flows associated with a project which have an initial investment cost equal to C_0. Internal rate of return (*IRR*) is calculated by setting.

$$\frac{R_1}{(1+r)}+\frac{R_2}{(1+r)^2}+\frac{R_3}{(1+r)^3}+\ldots\frac{R_n}{(1+r)^n}=C_0$$

or

$$\sum_{t=1}^{n}\frac{R_t}{(1+r)^t}=C_0$$

Solving for *r*, we get the internal rate of return.

Decision Rate : According to internal rate of return (*IRR*) method, *if the internal rate of return (r) of a capital project is greater than the cost of capital, investment in the project should be made.* If the internal rate of return (*r*) is less than the cost of capital, the project should be rejected. When the internal rate of return from a project exceeds the cost of capital, it will increase the value of the firm and is therefore, worthwhile to make investment in the project. For instance, if the use of funds costs 12 per cent per annum (that is, cost of capital is 12 per cent per, annum) to a firm and if internal rate of return from the project is 15 per cent, the project should be accepted for investment.

It may however be noted that it is not easy to calculate internal rate of return when the life of a project exceeds one year. Fortunately, these days electronic calculators are available with which we can easily calculate internal rate of return from the data. Further, calculations of present values and internal rate of return can be done using personal computers through simple computer programmes such as *Lotus 1-2-3* or *Microsoft Excel.*

However, in the absence of calculators and computers, internal rate of return (*IRR*) can be obtained through *trial and error method.* In this trial and error method, a discount rate is arbitrarily selected and the present value of cash flows generated by the project is calculated using the discount rate. If the present value so obtained is higher than the cost of capital to the firm, then a higher discount rate is chosen to evaluate the future cash flows of the project. If with this new higher discount rate, the present value of future cash flows of the project is lower than the initial cost of the project, then in that case repeat the process by reducing the discount rate. This iterative process should be continued until we find the discount rate which make present value of cash flows equal to the cost of the project. *This discount rate at which present value of future cash flows from a project becomes equal to the initial cost of the project to the firm represents the internal rate of return from the capital project*.

Let us illustrate it. *Suppose a firm is considering to invest in a project whose initial investment cost to it equals Rs. 1,000. Net cash flow from this project which start coming after a year is Rs. 450 per year for five years. After 5 years the project machine has no salvage value. Firm has to borrow investment funds at 20 per cent per annum from a bank. Calculate internal rate of return from the project and give advise whether to accept or reject the project.*

Solution :

$$\frac{R_1}{(a+r)}+\frac{R_2}{(a+r)^2}+\frac{R_3}{(1+r)^3}+\frac{R_4}{(1+r)^4}+\frac{R_5}{(1+r)^5}=C_0$$

$$\frac{450}{(1+r)}+\frac{450}{(1+r)^2}+\frac{450}{(1+r)^3}+\frac{450}{(1+r)^4}+\frac{450}{(1+r)^5}=1,000$$

Using the calculator *r* has been found to be equal to 34.9. Thus, 34.9 per cent is the internal rate of return from the investment in the proposed project. Since, internal rate of return (34.9%) exceeds

the cost of capital (*i.e.* cost of borrowed funds) of 20 per cent, the project should be accepted for investment.

Choice Among Several Investment Projects. The method of internal rate of return can also be used to make a choice among several projects. This issue is faced when a firm has limited funds for making investment. To decide about choosing among alternative investment projects on the basis of the criterion of internal rate of return, internal rates of return (*IRR*) are calculated from the cash flows of various available projects. Then, in terms of the magnitudes of internal rate of return of the various projects can be given rank-ordering, from the highest rate of return to the lowest rate of return. Then investment funds should be employed to those projects which yield highest possible rates of return from the given limited available funds.

Comparison of NPV and IRR Methods

When evaluating a single project, the NPV and IRR methods yield the same results. When net present value (NPV) of a project is positive, internal rate of return exceeds the cost of capital and when net present value (NPV) of a project is negative, internal rate of return is less than the cost of capital. This is because when net present value of a project is positive, it implies that the rate at which future cash flows have been discounted to obtain the net present value is greater than the implied rate of return which represents the discount rate which makes the present value of future net cash flows equal to the initial cost of project [2]. In fact, in case of a single project, NPV and IRR are the two sides of the same coin. Therefore, in case of a single investment project, the two evaluation methods always lead to the same accept-reject decision.

However, in case of two or more mutually exclusive projects, that is, when only one out of two or more projects has to be undertaken for investment, the two evaluation criteria can lead to contradictory results, Consider Table 45.3 where values of future net cash flows generated by two projects *A* and *B* are given, and both the net present value (NPV) and internal rate of return (IRR) of the two projects have been calculated. It will be seen from the table that the project *A* has lower net present value (Rs. 622) but a higher rate of return (34.9%), whereas project *B* has a higher net present value (675) but a lower internal rate of return (24.2). Thus, on the basis of the net present value (NPV) criterion project *B* should be chosen, while according to internal rate of return (IRR) criterion, project *A* is recommended for investment. So in this case we get inconsistent results from the two evaluation criteria. *The inconsistent results are due to the difference in the rate of return at which annual cash flows are reinvested for the next years*.

Whereas under the net present value method, the net cash flows generated by a project are implicitly assumed to be *reinvested at the firm's cost of capital used by the firm for evaluation*, in the IRR method the net cash flows generated by the project are *implicitly assumed to be reinvested at the same higher rate of return earned on the project*. This is somewhat unrealistic assumption because the rate of return on firm's reinvestment of its annual cash flows may not be as high as the cost of capital of the investment in the project under consideration. Therefore, *the net present value (NPV) is considered to be superior or better than internal rate of return (IRR) criterion in deciding which of the two mutually exclusive projects should be chosen for investment.* To conclude, when the two or more projects provides contradictory or inconsistent results on the basis of the two criteria, it is preferable for the firm to choose a project with a higher net present value (NPV) than the one with higher internal rate of return (IRR).

2. Note that internal rate of return can also be defined as discount rate which makes net present value (NPV) of the project equal to zero.

Table 45.3 Comparing NPV and IRR Methods

Net Cash Flows

Years	*0**	*1*	*2*	*3*	*4*	*5*	*NPV*	*IRR*
Project *A*	1,000	450	450	450	450	450	622	34.9
Project *B*	1,000	– 300	0	600	600	2,000	675	24.2

** Initial Investment Outlay (i.e., Cost of the Project) in Year 0.*

Profitability Index (PI) Method

Another criterion of evaluating investment projects, which is similar to the net present value method, is that of profitability index which is also called *benefit-cost ratio.* Profitability index (PI) of a capital project is defined as the ratio of present value of future cash flows from a project to the initial cost of the project. Thus

$$\text{Profitability Index (PI)} = \frac{\text{Present Value of Future Cash Flows}}{\text{Initial Cost of a Project}}$$

or

$$\text{PI} = \sum_{t=1}^{n} \frac{R_t}{(1+i)^t}$$

where $\sum_{t=1}^{n} \frac{R_t}{(1+i)^t}$ is the sum of the present value of future cash flows from the project and C_0 is the initial cost of a project.

When profitability index of a project is greater than one, it means it adds more to the value of a firm it costs.

Decision rule. It is clear from above that in the method of profitability index, *a project with profitability index greater than one* (*PI* > 1) *should be accepted and a project with a profitability index less than one (PI < 1) should be rejected.* This means that those projects are accepted for investment which yield more than a rupee of discounted each flows for each rupee cost. Profitability index and net present value criterion always lead to consistent results regarding accept/reject decision when a single independent project is being evaluated. This is because when profitability index is greater than one (*PI* > 1), it implies net present value is positive (NPV > 0) and when profitability index is less than one, it implies not present value is negative (NPV < 0).

However, it may be noted that when choice among alternative projects of unequal size (that is, requiring different amounts of investment outlay) are concerned, the NPV and profitability index (*PI*) can provide different rankings to the various projects.

The Level of Investment Expenditure and Capital Rationing

Two important decisions are required to be made in the capital budgeting process of a firm. First, if enough funds are available for investment with a firm, it has to determine the level of capital expenditure it will make. Secondly, if the limited amount of funds are available, it has to decide how a firm should allocate them among various projects. Whereas the first type of decision relates to the *determination of the capital expenditure,* the second type of decision-making is called *capital rationing.*

Let us explain what determines the level of capital expenditure by a firm when there is no constraint on the resources available to it. The level of capital expenditure depends on the *internal rate of return* (IRR) of the project on the one hand and *capital expenditure* (*i.e.* required funds for investment) on the other. In Fig. 45.3 we have drawn histograms representing seven projects. The length of a histogram shows the internal rate of return (IRR) of a project, while the width of a

histogram represents the cost of the project, that is, funds initially required to implement it.

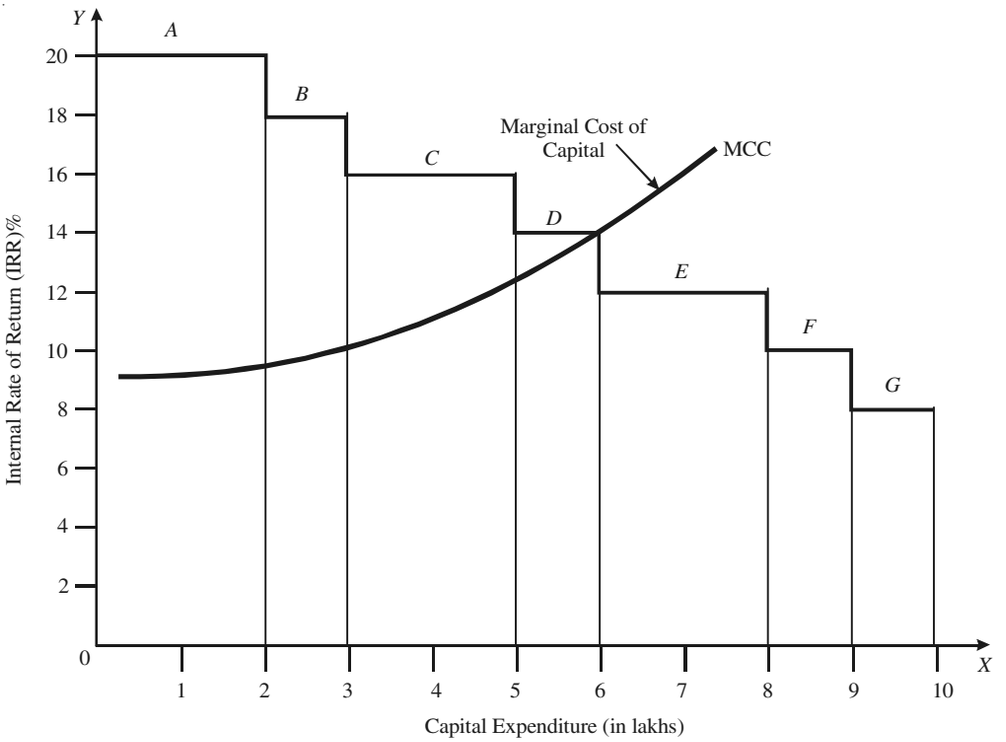


Fig. 45.3 *Determining Capital Expenditure*

The various projects are ranked in order of internal rate of return from them. Thus IRR of project *A* is 20%, of project *B* is 18%, of project *C* is 16% and so on. The thick line on the top of the histograms indicates decreasing rate of return of successive investment projects.

MCC curve shows the marginal cost of capital which is shown to be increasing. This is because as the firm borrows more funds, the greater the degree of risk borne by the lenders to the firm. This is because as the firm borrows more funds, there is a greater risk of default in its paying back the funds borrowed. Therefore, as a firm borrows more funds a higher rate of interest will be demanded by the lenders. *A project will be undertaken if internal rate return exceeds marginal cost of capital.* Thus, a firm will invest in new projects as long as internal rate of return (*IRR*) from investment projects is greater than the cost of capital. It will be seen from Fig. 45.3 that given the cost of capital as shown by the curve *MCC*, the firm will undertake investment projects *A, B, C* and *D.* It will not go further as internal rate of return (IRR) on additional projects is less than the cost of capital. It will be seen from Fig. 45.3 that the firm will make total investment or incur capital expenditure equal to Rs. 6 lakhs.

Capital Rationing. However, unconstrained investment funds for a firm are not generally available. Quite often a firm has a limited amount of investment funds. With the given constraint of available funds, it has to decide which projects it should choose for implementation and accordingly allocate or ration capital funds to them.

Suppose investment funds available for a firm for investment are Rs. 4 lakhs. It will be seen from the Table 45.4 that projects *A, B* and *D* cost Rs. 4 lakhs ; and projects *A* and *C* also together cost Rs. 4 lakhs; and projects *A* and *B* (which together cost 3 lakhs and therefore leave Rs. one lakh unutilised). In this case of choice of projects with a limited resources, the use of criteria of NPV and IRR conflict

with each other. This will be easily understood from the data given in Table 45.4

Table 45.4 Capital Rationing

Project	Initial Cost of Project	NPV	IRR	IRR (Raking)
A	2,000,00	10,000	20	1
B	1,000,00	4,000	18	2
C	2,000,00	7,000	16	3
D	1,000,00	2,500	14	4
E	1,000,00	2,000	13	5
F	1,000,00	1,900	11	6
Idle or Unutilised Balance				

In this case *optimum choice is one which provides the highest* NPV *to the firm.* It will be seen from the table that the sum of NPV of the projects *A, B* and *D* are Rs. 16,500 (10,000 + 4,000 + 2,500 = Rs. 16,500), the sum of NPV of the projects *A* and *C* is Rs. 17,000 (10,000 + 7,000 = 17,000) and NPV of the projects *A* and *B* (with idle funds of Rs. one lakh is 10,000 + 4,000 + 0 = 14,000). Thus, the *NPV* of the projects *A* and *C* is the maximum. This means that if Rs. 4 lakhs are invested in them, they will increase the value of the firm by the largest amount among the alternative choices.

It should be carefully noted that the ranking of internal rate of return (IRR) of project *C* comes third, while project *B* comes second but it is not chosen on the basis of the NPV criterion. For the choice of projects when a firm faces constraint of resources, the use of NPV criterion for choosing projects for investment ensures the maximum profitability.

INVESTMENT DECISION BY CONSUMERS

We have explained above how firms decide to invest in a project that lasts for years and generates a stream of profits for it. The firm decides on the basis of its estimate of the net present value of (NPV) of the project. The consumers also face a similar problem when they purchase a *durable consumer good* such as a car, air conditioner. Since these durable goods provide to a stream of future services or benefits to the consumer, he compares the value of the stream of services or benefits which a durable good provides him with its cost. Suppose a consumer is to decide whether to buy a new car. Since he will keep the car for a number of years, say 8 years, most of services or benefits from owning a car will accrue to him in future. He will also incur costs on insurance, maintenance and petrol on using a car for travelling. He will therefore evaluate the stream of net benefit after deducting these costs on maintenance, petrol etc. He will then decide whether to buy the car by comparing present the value of the stream of benefits from the car with the current purchase cost. Similarly, when a consumer has to decide about purchasing an air conditioner he will calculate the present value of the stream of future net benefits, that is, benefits obtained in the form of comforts of a cool room from the air conditioner less the cost of electricity used for operating it. *The decision to buy the car is determined by net present value (NPV) criterion as explained in case of a firm deciding to invest in a project.*

The benefit from porsessing a car is mainly of transportation services it provides to the owner. The value of these services a car provides differs from individual to individual. Suppose the value of services or benefits obtained by a consumer per year is denoted by *B* and cost incurred on operating it annually by *E* and after 8 years its sales value is Rs. 50,000, then the net present value (NPV) can be stated as under.

$$NPV = -C + B - E + \frac{B-E}{(1+r)} + \frac{B-E}{(1+r)^2} + ... + \frac{B-E}{(1+r)^8} + \frac{50,000}{(1+r)^8}$$

where C is the purchase cost of the car.

If the net present value of a car is positive, that is, if the consumer evaluates the flow of benefits which he expects from owning a car is greater than its purchase cost, he will buy the car.

Now at what rate the consumer will discount the future benefits he expects to receive from the car? As explained above in connection with the discount rate to be used by the firms who have to decide about capital investment the *appropriate discount rate is the opportunity cost of money*. If the consumer has to borrow money for buying the car, then the discount rate would be the market rate of interest at which he borrows the required funds. This rate is likely to be higher than the interest rate on bonds or fixed deposits in the banks. If the consumer uses its own funds to purchase the car, then the discount rate representing the opportunity cost of the funds is the return that he would earn by investing the funds in any alternative assets such as corporate bonds or fixed deposits of commercial banks

Optimal Durability of Durable Consumer Goods

Two aspects of the durable goods which the consumers buy are worth examining. First, what is the optimal durability of goods. Secondly, how a change in the discount rate affects durability of goods and how the change in the durability will affect the price of the product. To analyse the optimal durability we assume for the sake of simplicity that the goods provides a stream of services or benefits for a certain number of years and then breaks down and therefore has no salvage or scrap value. Since the firms producing durable goods have to incur extra costs to increase the durability of their products, that is, improving their quality so that they can last longer, the marginal cost to the firms of increasing the durability of the products will be increasing as shown by *TC* curve in Figure

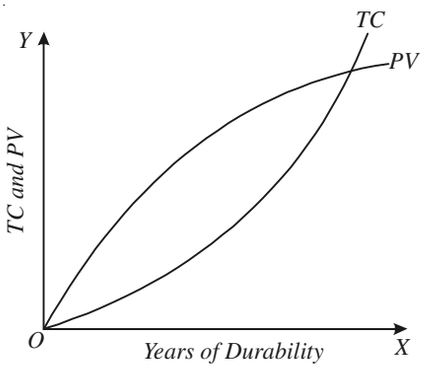


Fig. 45.4. *Impact of Increase in Durability on Total Cost and Present Value of a Durable Consumer Good*

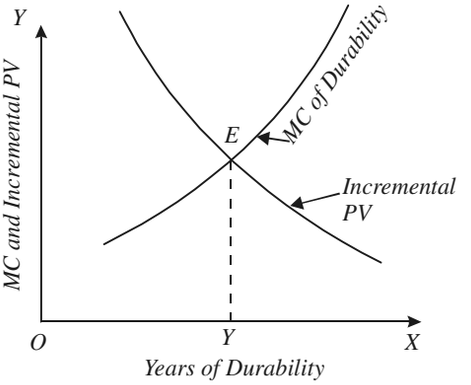


Fig. 45.5. *The Determination of Optimal Durability of a Good*

45.4. On the other hand, the the increment to present value of the good will decline as the years of durability of the good increases. Therefore, the slope of PV curve will be declining with the increase in the durability of the good as shown by the PV curve in Figure 45.4. Consistent with these total cost (*TC*) and PV curves of increasing durability of the good we have drawn the upward sloping *MC* curve of increases durability and downward-sloping incremental *PV* curve of the increase in durability of the good in Fig. 45.5. These *MC* and incremental *PV* curves of increasing durability intersects at point *E* and accordingly determines life of *OY* years as optimal durability of the consumer good.

Impact of Change in the Discount Rate on the Durability and Demand for the Good

How will the durability of the good and demand for it change as a result of change in rate of interest at which stream of future service benefits from it are discounted. Increase in the discount

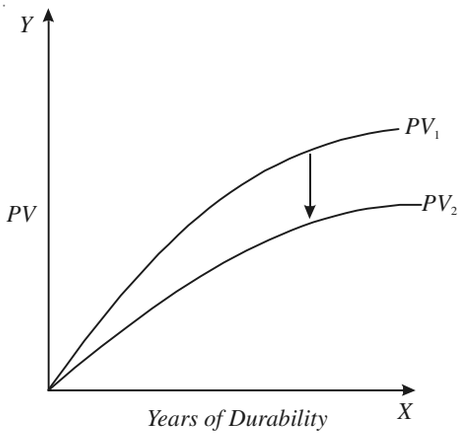


Fig 45.6.*Downward Shift in the PV Cruve as a Result of Increase in the Discount Rate*

rate will reduce the present value of an asset and will therefore cause a downward shift in the present value curve from PV_1 to PV_2 as shown in Figure 45.6. Besides, increase in the discount rate will also cause a fall in the incremental *PV* for each unit increase in the durability causing a downward shift in incremental *PV* curve as depicted in Fig. 45.7 It will be seen from Fig. 45.7 that increase in the discount rate reduces the optimal durability of a good from Y_1 to Y_2.

It follows from above that two factors account for the decline in the present value of benefits from a durable good as result of increase in the interest rate (*i.e.* the discount rate). First, the present value (*PV*) of a fixed stream of benefits decreases as the interest rate at which future benefits are discounted increases. Second, the durability of a good declines as the discount rate increases

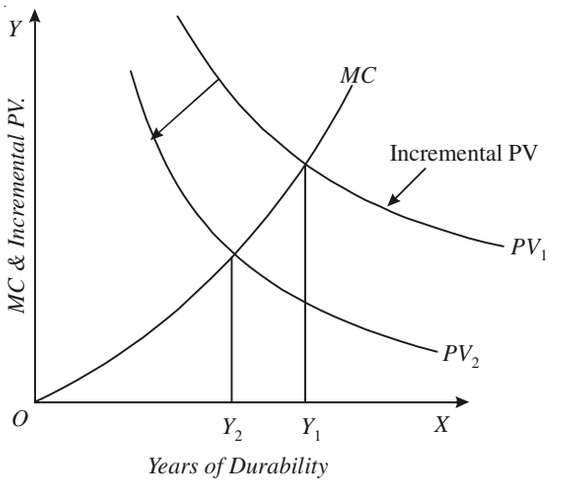


Fig. 45.7. *Increase in the Discount Rate Reduces Optimal Durability.*

causing a reduction in the *length of the stream of future benefits* from a durable good. Since the price which a consumer is willing to pay for the good is determined by the present value of the stream of service benefits from a good received by him *the increase in the discount rate reduces the demand for the durable good and therefore its price*.

DETERMINATION OF INTEREST : A COMPLETE ANALYSIS

In the previous chapter we have explained the determination of rate of interest first through borrowing-lending equilibrium in a pure exchange model when net saving was zero and then we incorporated positive saving and investment in our analysis of the determination of rate of interest. We are now in a position to explain the complete model of determination of rate of interest. As is well known *interest rate is the price at which funds are borrowed*. Therefore, rate of interest is determined by demand for and supply of loanable funds. The supply of loanable is provided by households who save out of their current income so as to spend more in future. Some persons earn high incomes during their working age but after retirement their incomes become very much less. Saving in the years of their service allow them to spread their income more evenly over time. Besides, since they earn interest on their savings they are able to consume more in future for each sacrifice of present consumption. As a result, the higher the interest rate, the households are induced to save more and lend more. The supply curve of loanable funds therefore slopes upward to the right as shown by *SL* curve in Figure 45.8

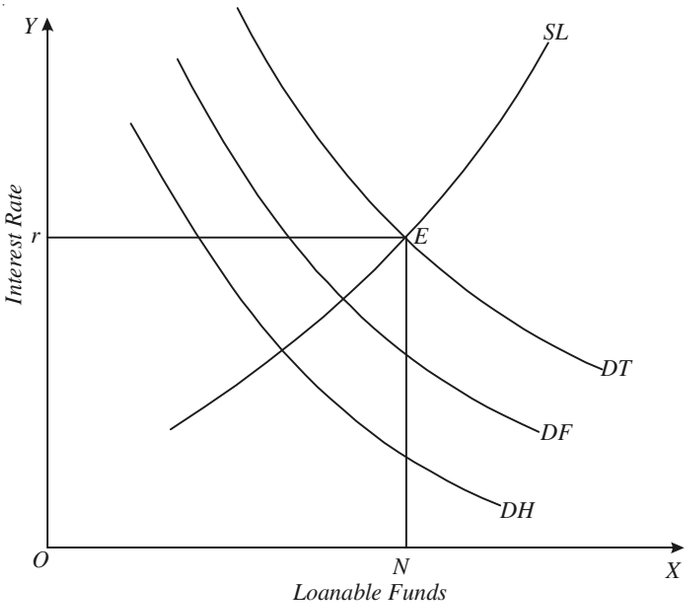


Fig. 45.8 *Determination of Rate of Interest through Demand for and Supply of Loanable funds*.

The demand for loanable funds mainly comes from two sources : households and firms. Those households who want to consume more than their income which are currently small but are expected to increase in future demand loanable funds. Those households who plan to buy expensive items such as houses and cars for which they want to make payment out of their future incomes also demand funds from the market. The higher the interest rate, the greater the cost of borrowing funds from the market. Therefore, the demand curve *DH* of loanable funds by households in Fig. 45.8 is sloping downward.

The business firms who want to make capital investment also demand funds from the market. As explained earlier, the firms invest in those projects whose net present value (NPV) are positive which implies that expected yields from the projects are greater than the opportunity cost of funds. And the opportunity cost in most cases is the market rate of interest adjusted for risk involved. Business firms borrow loanable funds from the market as the return from investment will accrue to them in future while the payment for the investment project has to be made at present. The demand for funds for investments by business firms will be less as the cost of borrowing (*i.e.* rate of interest) is higher

and vice versa. Therefore, the demand curve of loanable funds for investments purposes by the firms slopes downward as is shown by the *DF* curve in Figure 45.8.

The total demand curve for loanable funds is obtained by summing up the demands for loanable funds both by the households and the business firms. Therefore, by making horizontal summation of demand curves of loanable funds *DH* of households and *DF* of firms we get the total market demand curve *DL* of loanable funds in Figure 45.8.

The intersection of the total demand curve for loanable funds *DL* and the supply curve of loanable funds *SL* in Fig. 45.8 determine equilibrium interest rate *r* and the funds borrowed and supplied are equal to *ON*.

Changes in Equilibrium Interest Rate

Our analysis also shows what can bring about changes in equilibrium interest rate. The changes in interest rate can occur because of changes in either the supply of loanable funds or the demand for loanable funds. First, take the demand side of loanable funds. When there is a depression in the economy, the expected sales of the products in the future will decrease causing lower stream of profits from new investment projects. This will reduce their NPV and the firms will be less willing to undertake investment in new projects. This will bring about a shift in the demand curve *DF* of loanable funds by the firms to the left. This will tend to reduce equilibrium rate of interest. On the other hand, if there is technological improvement that raises the productivity of capital. Given the other things such as price of the products, rise in productivity of capital will raise the stream of future profits which will tend to increase investment and lead to the increase in demand for loanable funds by the firms. This will work to raise the equilibrium interest rate.

Besides, in our analysis of determination of rate of interest we have ignored the role of the government. In the modern times, government throughout the world also borrow funds to meet their fiscal deficits. For example, in India for the last several years the government is running huge fiscal deficits which have been largely financed by borrowing funds from the market. The increase in government demand for funds will shift the demand curve *DL* for loanable funds to the right and will therefore tend to raise the equilibrium interest rate.

The equilibrium rate of interest can also change as a result of shift in the supply curve of loanable fund. If there is a general shift in individuals' preferences in favour of current consumption over future consumption, then while, on the one hand, the supply curve of funds to lend money will shift to the left, the demand curve for borrowed funds will increase which will shift the demand curve for loanable funds to the right. The effect on the rate of interest will depend on which of these two effects is greater.

Finally, the monetaring policy of the Central Bank of the country also affects the rate of interest. When Central Bank adopts an expansionary monetary policy and through open market operations purchases the securities, it increases the supply of lendable funds by the banks. As a result, supply of loanable funds increases causing a shift in the supply curve *SL* of loanable funds to the right and lowers the market rate of interest. On the other hand, if the Central Bank adopts a tight monetary policy and sells securities in the market, it can mop up liquidity in the market and reduce the supply of lendable funds of the commercial banks. This will tend to raise the market interest. The Central Bank of the country (*i.e.* RBI in India) can also influence the market rate of interest by changing the Cash Reserve Ratio (CRR) of the banks and thereby affect money supply and rate of interest in the economy. Thus monetary policy of the Central Bank is also an important determinant of rate of interest. It is important to note that in our basic model of determination of market rate of interest we have ignored the role of government and the Central Bank of the country in the determination of market rate of interest in the economy.

QUESTIONS FOR REVIEW

1. Explain the rule for calculating present discounted value of a perpetual income stream. What is the worth of a perpetuity paying Rs. 10,000 per year when the rate of interest is 8 percent per annum *D.U.BA(Hons.), 1986*

2. Given the following three projects, which project has the maximum net discounted present value assuming the market rate of interest as 8%. *D.U. BA(Hons.), 1987*
   (*i*) Project *A* which yields an income of Rs. 200 per year indefinitely into the future.
   (*ii*) Project *B* which yields Rs. 400 per year for two years
   (*iii*) Project *C* which yields Rs. 100 today and Rs. 200 after 2 years

3. For a certain project the anticipated cash flows are Rs. 100 at present and Rs. 125 next year. What is the present value when the interest rate is 10% ? What is the highest rate of interest at which the project should be undertaken ?

4. Calculate the market equilibrium value for the following assets. Assume the market rate of interest is 10%.
   (*i*) A bond that pays Rs. 100 interest per year for two years and is paid off at Rs. 1000 at the end of two years
   (*ii*) A building that will earn Rs. 100 per year for two years and then will earn Rs. 50 per year indefinitely into the future.

5. What is the net present value criterion for capital investment decision ? Explain *D.U., B. Com. (Hons.) 2006*

6. What is the internal rate of return on an investment ? Calculate the internal rate of return on an investment yielding Rs. 25,000 after one year and costing Rs. 20,000. Suppose instead that this investment yielded nothing in the first year and Rs. 30,000 after two years, what would be its internal rate of return ? *D.U. BA (Hons.), 1997*

7. An investment pays Rs. 100 per year for ever starting a year from now what is its present value if the rate of interest is 10 per cent ? What is its present value if the rate of interest is 20 per cent? *D.U. BA (Hons.) 2000*

8. Calculate the market value for an acre of land yielding Rs. 100 per annum, after all expenses, indefinitely into the future. Market rate of interest is 10%.
(**Hints :** This is a case of perpetuity) *D.U. BA (Hons.) 1997*

9. Suppose the interest rate is 10 per cent. What is the value of a coupon bond that pays Rs. 80 per year for each of the next five years and then makes principal repayment of Rs. 1000 in the sixth year? What is the value of the bond if interest rate is 15 per cent?

10. What is meant by the effective yield on a bond ? How does one calculate it? Why do some corporate bonds have effective yields that are higher than others.

11. A bond has two years to mature. It makes coupon payment of Rs. 100 after one year and both a coupon payment of Rs. 100 and principal repayment of Rs. 1000 after two years. The bond is selling for Rs. 966. What is its effective yield ?

12. How do investors calculate the present value of a bond ? What do you mean by effective yield on bond ? If the interest rate is 5% per annum, what is the present value of a perpetuity that pays Rs. 1,000 per year for ever ? *D.U. B.Com (Hons.) 2006, 2009*

13. What is the net present value (NPV) criterion for investment decision? How is net present value of investment projects calculated? If all the cash flows for the project are certain, what discount rate should be used to calculate the NPV.

14. How is the present value of a bond calculated ? If the interest rate is 5 per cent, what is the present value of a perpetuity that pays Rs. 1000 per year for ever ?

15. Why does an investor need to discount when he gets cash flows in future ? Suppose an investment A entitles an investor perpetual income stream of Rs. 1,000 per annum, it costs him Rs., 10,000 now. Another investment B costs him Rs. 3,5000 now and entitles him to an income stream of Rs. 1,100 after one year, Rs. 2,420 after two years and Rs. 1,331 after three years. Which of these two would you advise him to go for ? Assume that the rate of interest is 10% per annum for the period. *D.U., B.Com (Hons.) II year 2009*

16. What do you mean by loanable funds ? How do their demand and supply determine interest rate ? *D.U., B.Com (hons) II Year 2007, 2009*

17. What are the factors that a consumer takes into account when deciding to buy or not to buy a car (or air conditioner) *D.U., BCom (Hons.) II Year*

[**Hint:** For answer see section on Investment Decision by Consumers of the preceding Chapter 45.]

APPENDIX TO CHAPTER 45

Table I : Present Value of Re 1.00

$$PVIF_m = \frac{1}{(1+i)^n}$$

Period	Interest (discount(Rate (i)										Interest					
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	11%	12%	14%	15%	16%		
1	.9901	.9804.	.9709	.9615	.9524	.9434	.9346	.9259	.9174	.9634	.8929	.8772	.8696	.8621		
2	.9803	.9612	.9426	.9246	.9070	.8900	.8734	.8573	.8417	.8364	.7972	.7695	.7561	.7432		
3	.9706	.9423	.9151	.8890	.8638	.8396	.8163	.7938	.7722	.7583	.7118	.6750	.6575	.6407		
4	.9610	.9238	.8885	.8548	.8227	.7921	.7629	.7350	.7084	.6832	.6355	.5921	.5718	.5523		
5	.9215	.9057	.8626	.8219	.7835	.7473	.7130	.6806	.6499	.6219	.5674	.5194	.4972	.4761		
6	.9420	.8880	.8375	.7903	.7462	.7050	.6663	.6302	.5963	.5645	.5066	.4556	.4323	.4104		
7	.9327	.8706	.8131	.7599	.7107	.6651	.6227	.5835	.5470	.5132	.4523	.3996	.3759	.3538		
8	.9235	.8536	.7894	.7307	.6768	.6274	.5820	.5403	.5019	.4565	.4039	.3506	.3269	.3050		
9	.9143	.8368	.7664	.7026	.6446	.5919	.5439	.5002	.4604	.4241	.3606	.3075	.2843	.2630		
10	.9053	.8203	.7441	.6756	.6139	.5584	.5083	.4632	.4224	.3855	.3220	.2697	.2472	.2267		
11	.8963	.8043	.7224	.6496	.5847	.5268	.4751	.4289	.3875	.3505	.2875	.2366	.2149	.1954		
12	.8874	.7885	.7014	.6246	.5568	.4970	.4440	.3971	.3555	.3156	.2567	.2076	.1869	.1685		
13	.8787	.7730	.6810	.6606	.5303	.4688	.4150	.3677	.3262	.2897	.2292	.1821	.1625	.1452		
14	.8700	.7579	.6611	.5775	.5051	.4423	.3878	.3405	.2992	.2533	.2046	.1597	.1413	.1252		
15	.8613	.7430	.6419	.5553	.4810	.4173	.3624	.3152	.2745	.2394	.1827	.1401	.1229	.1079		
16	.8528	.7284	.6232	.5339	.4581	.3936	.3387	.2919	.2519	.2176	.1631	.1229	.1069	.0930		
17	.8444	.7142	.6050	.5134	.4363	.3714	.3166	.2703	.2311	.1975	.1456	.1078	.0929	.0802		
18	.8360	.7002	.5874	.4936	.4155	.3503	.2959	.2502	.2120	.1799	.1300	.0946	.0808	.0691		
19	.8277	.6864	.5703	.4746	.3957	.3305	.2765	.2317	.1945	.1635	.1161	.0829	.0703	.0596		
20	.8195	.6730	.5537	.4564	.3769	.3118	.2584	.2145	.1784	.1436	.1037	.0728	.0611	.0514		
21	.8114	.6598	.5375	.4388	.3589	.2942	.2415	.1987	.1637	.1351	.0926	.0638	.0531	.0443		
22	.8034	.6468	.5219	.4220	.3418	.2775	.2257	.1839	.1502	.1228	.0826	.0560	.0462	.0382		
23	.7954	.6342	.5067	.4057	.3256	.2618	.2109	.1703	.1378	.1117	.0738	.0491	.0402	.0329		
24	.7876	.6217	.4919	.3901	.3101	.2470	.1971	.1577	.1264	.1015	.0659	.0431	.0349	.0284		
25	.7798	.6095	.4776	.3751	.2953	.2330	.1842	.1460	.1160	.0923	.0588	.0378	.0304	.0245		
26	.7720	.5976	.4637	.3607	.2812	.2198	.1722	.1352	.1064	.0839	.0525	.0331	.0264	.0211		
27	.7644	.5859	.4502	.3468	.2678	.2074	.1609	.1252	.0976	.0763	.0469	.0291	.0230	.0182		
28	.7568	.5744	.4371	.3335	.2551	.1956	.1504	.1159	.0895	.0693	.0419	.0255	.0200	.0157		
29	.7493	.5631	.4243	.3207	.2429	.1846	.1406	.1073	.0822	.0630	.0374	.0224	.0174	.0135		
30	.7419	.5521	.4120	.3083	.2314	.1741	.1314	.0994	.0754	.0573	.0334	.0196	.0151	.0116		
35	.7059	.5000	.3554	.2534	.1813	.1301	.0937	.0676	.0490	.0356	.0189	.0102	.0075	.0055		
40	.6717	.4529	.3066	.2083	.1420	.0972	.0668	.0460	.0318	.0221	.0107	.0053	.0037	.0026		
45	.6391	.4102	.2644	.1712	.1113	.0727	.0476	.0313	.0207	.0137	.0061	.0027	.0019	.0013		
50	.6080	.3715	.2281	.1407	.0872	.0543	.0339	.0213	.0134	.0085	.0035	.0014	.0009	.0006		
55	.5785	.3365	.1968	.1157	.0683	.0406	.0242	.0145	.0087	.0053	.0020	.0007	.0005	.0003		